Policy gradient methods

Lecture 2: baselines, actor-critic, GAE and PPO

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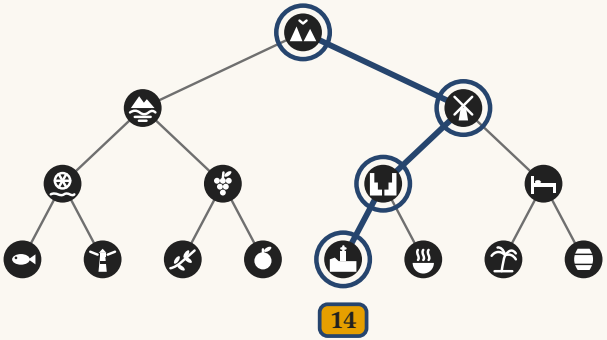
Part 4

REINFORCE with a baseline

Adds. A new multiplier, $\Psi_t = G_t - V(s_t)$: the coins of a path compared with the average path from the same place.

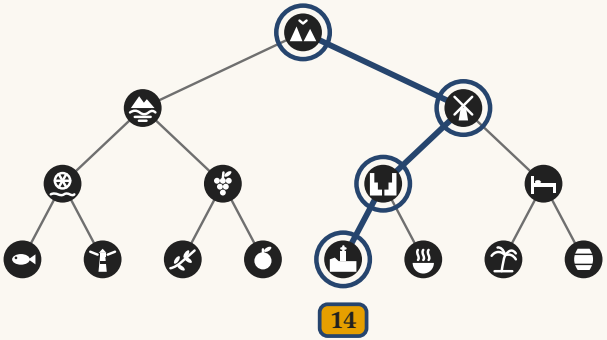
Answers. G_t alone does not say whether a path is good, and every road of every day is pushed up, which limits the step REINFORCE can take.

G_t alone is not very informative



1 One day on the outlined path paid $G_1 = 14$ coins, luck included.

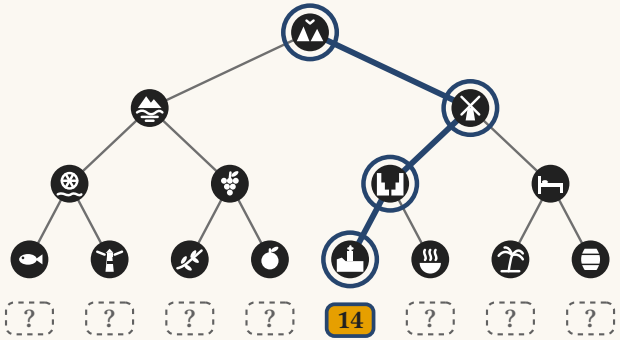
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2 Is this path good? Is it bad?

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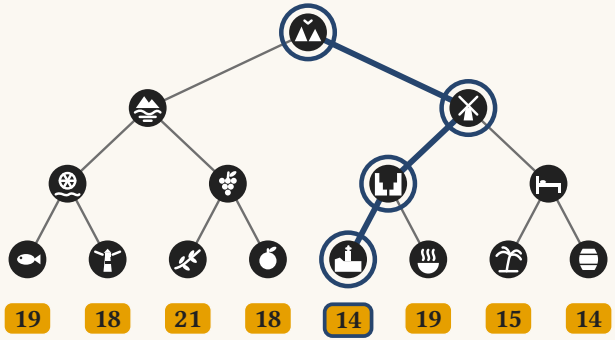


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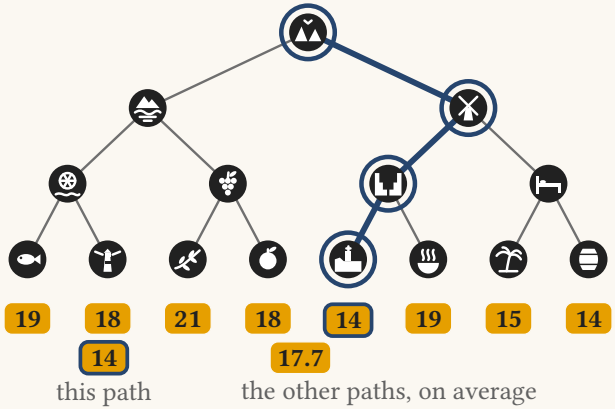
3 It depends on what the other paths yield.

If the other paths pay more, this path is poor



1 The gold numbers are what a day on each path pays on average.

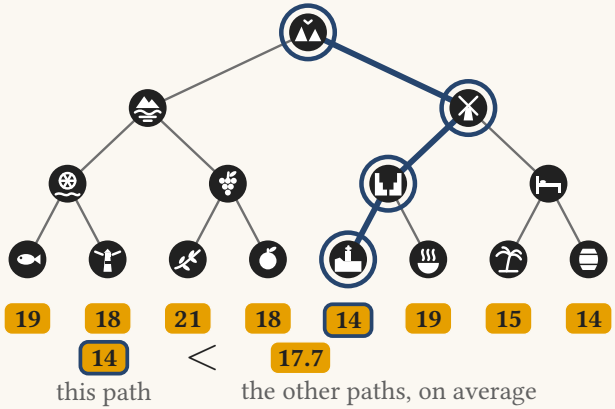
If the other paths pay more, this path is poor



1 The gold numbers are what a day on each path pays on average.

2 The other 7 paths average 17.7 coins, more than the 14 of this day.

If the other paths pay more, this path is poor

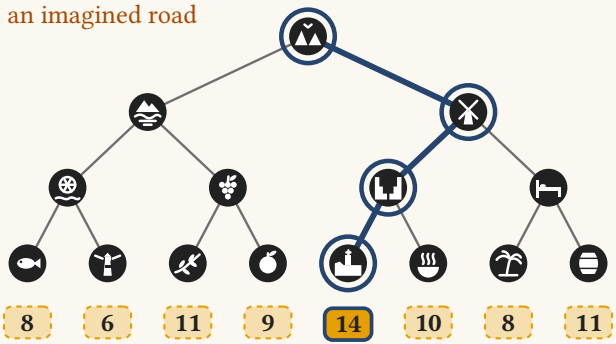


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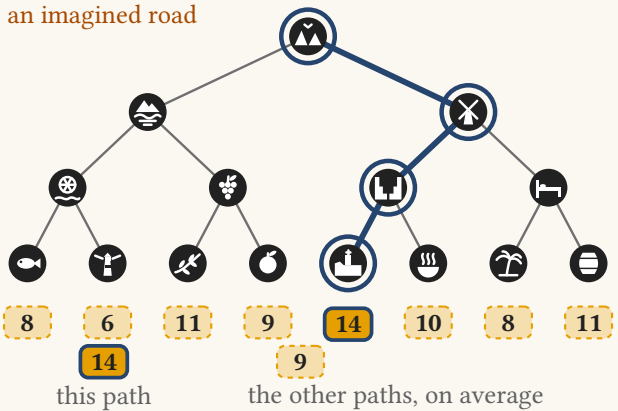
3 Next to them, this path is poor.

If the other paths pay less, this path is good



1 On an imagined road the other paths would pay the dashed numbers.

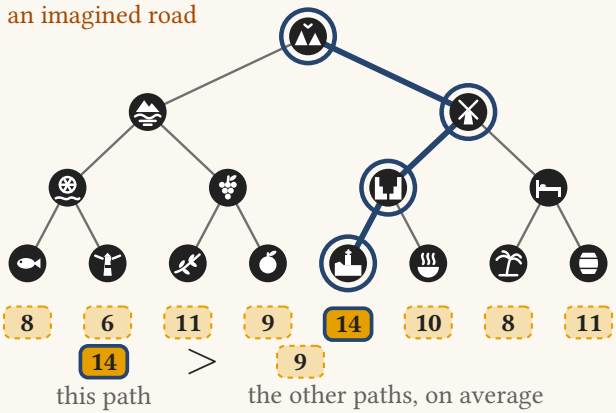
If the other paths pay less, this path is good



1 On an imagined road the other paths would pay the dashed numbers.

2 The other 7 paths average 9 coins, less than the 14 of this day.

If the other paths pay less, this path is good



1 On an imagined road the other paths would pay the dashed numbers.

2 The other 7 paths average 9 coins, less than the 14 of this day.

3 Next to them, the same path is good. The multiplier should include what the other paths yield.

The multiplier compares a path with the average

$$\Psi_t = G_t$$

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$$\cancel{\Psi_t = G_t} \longrightarrow \Psi_t = G_t - \boxed{V(s_t)}$$

1. Part 3's multiplier is $\Psi_t = G_t$. Coins alone do not say whether a path is good.
2. The new multiplier subtracts $V(s_t)$. A number subtracted like this is a baseline.

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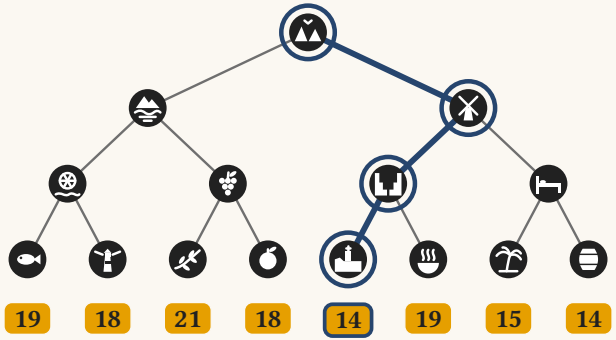
the average of G_t over all the paths from s_t

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2 The new multiplier subtracts $V(s_t)$. A number subtracted like this is a baseline.

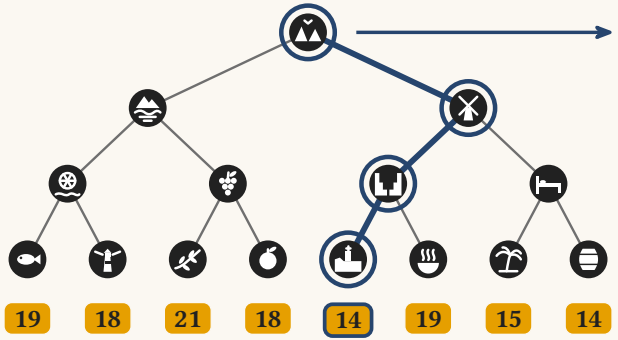
3 $V(s_t)$ averages G_t over all the paths that π_w can take from s_t, each weighted by its chance, the path itself included.

Example: the uniform policy gives $V(s_1) = 17.25$



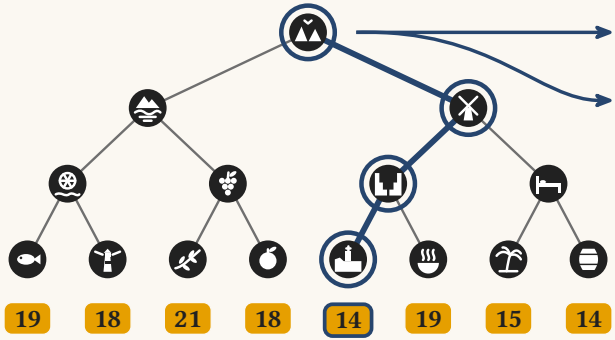
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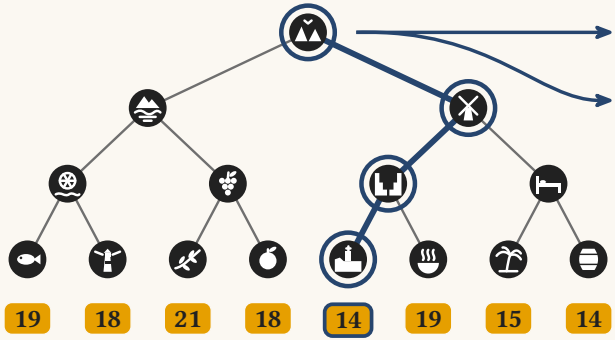


$G_1 = 14$

$V(s_1)$: the average of G_1 over all 8 paths

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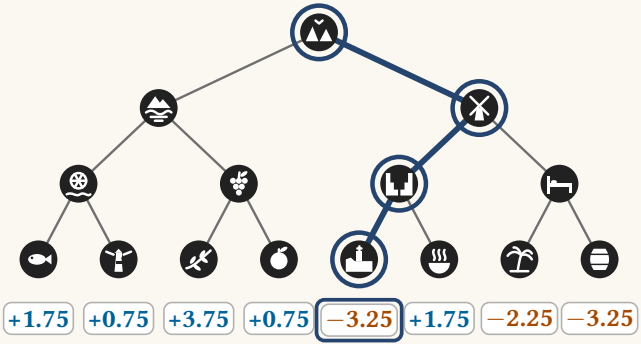
$G_1 = 14$

$V(s_1)$: the average of G_1 over all 8 paths

1. The uniform policy: every road has chance 1/2, so each path has chance 1/8.
2. The outlined path is one of the eight. The other 7 average 17.7.

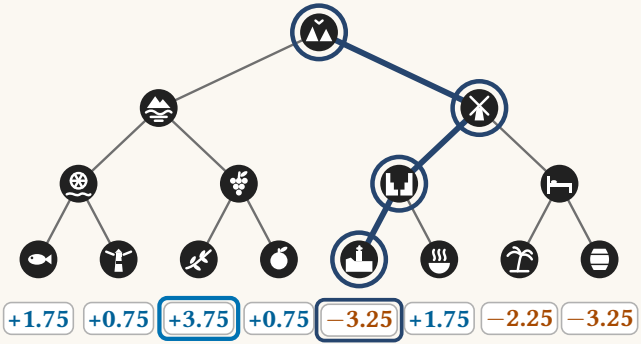
$$V(s_1) = (19 + 18 + 21 + 18 + 14 + 19 + 15 + 14) \,/\, 8 = 17.25$$

The best path gets the largest positive multiplier



1 For each path, $\Psi_1 = G_1 - V(s_1)$. For the outlined path, $14 - 17.25 = -3.25$.

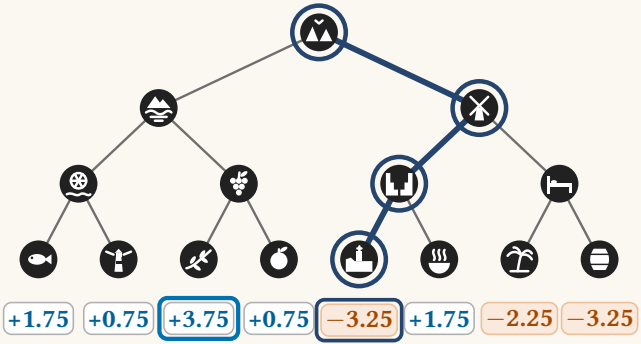
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2 The best path, 21 coins, gets the largest multiplier, +3.75.

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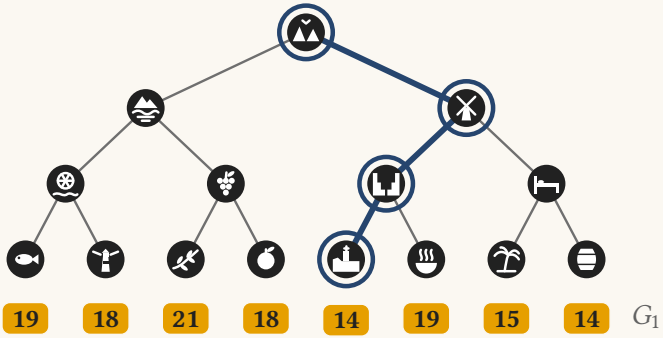


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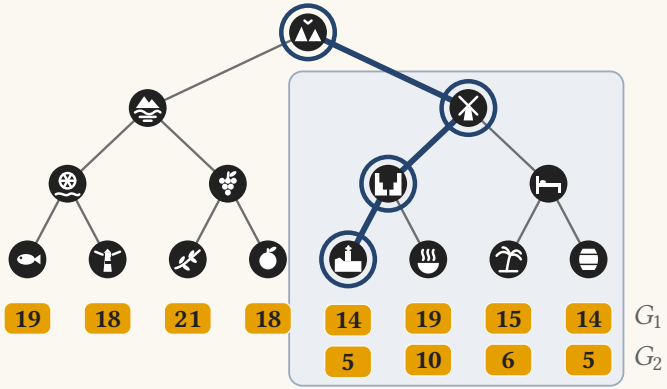
3 The three paths below the average get a negative multiplier.

$V(s_t)$ depends on where Mara stands



1 $V(s_t)$ averages G_t over the paths still ahead. At step 1 all 8 are: $V(s_1) = 17.25$.

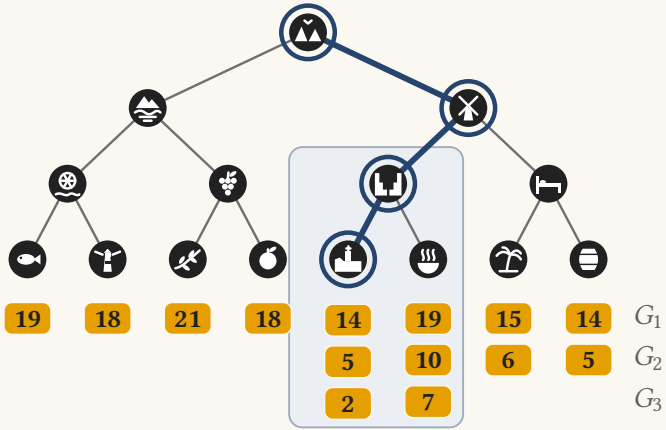
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2 At step 2 Mara stands at the market and 4 paths remain: $V(s_2) = 6.5$.

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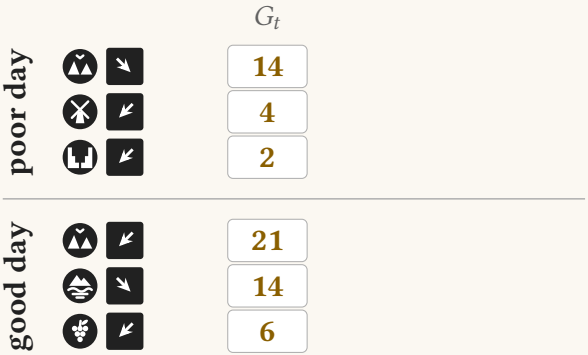


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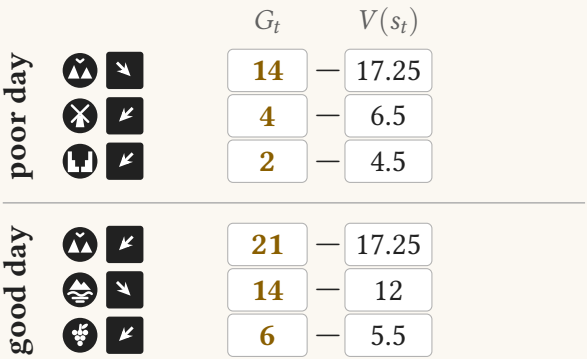
3 At step 3 she stands at the canyon and 2 remain: $V(s_3) = 4.5$.

The poor day now pushes its roads down



1 Part 3 pushed every road by G_t: even the poor day pushes its three roads up.

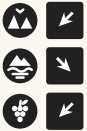
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The poor day now pushes its roads down

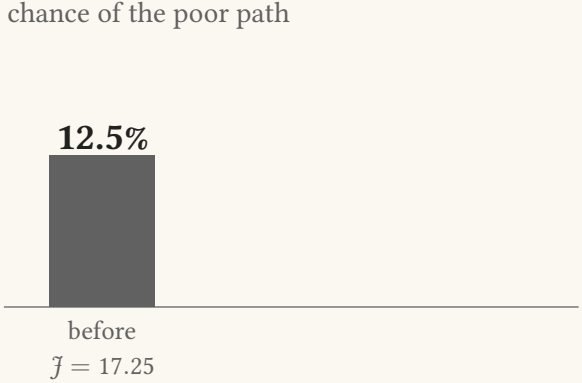
		G_t		$V(s_t)$		Ψ_t
poor day		14	−	17.25	=	−3.25
		4	−	6.5	=	−2.5
		2	−	4.5	=	−2.5
good day		21	−	17.25	=	+3.75
		14	−	12	=	+2
		6	−	5.5	=	+0.5

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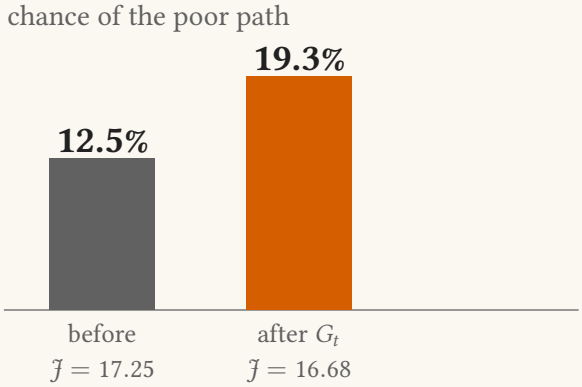
3 The new push is $G_t - V(s_t)$: the poor day's roads go down, the good day's stay up.

With the baseline the poor path becomes rarer



1 From the uniform table, update the knobs with the poor day alone, with step size 0.05.

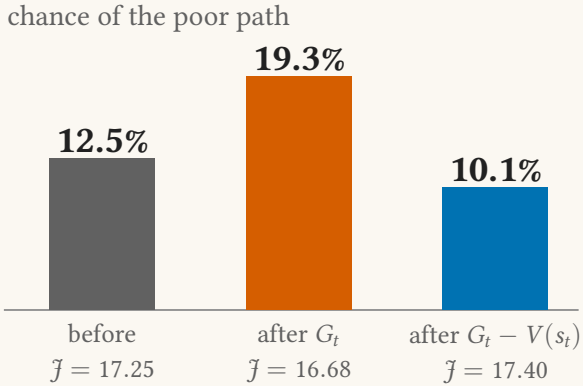
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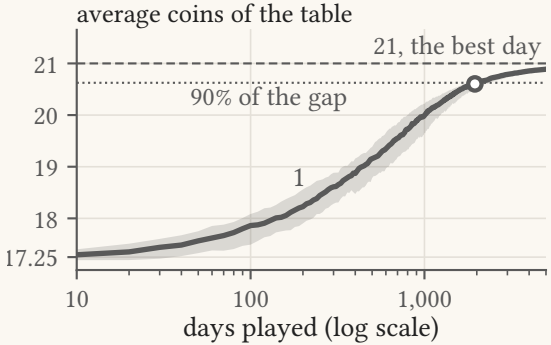


1 From the uniform table, update the knobs with the poor day alone, with step size 0.05.

2 With G_t the poor path becomes likelier and $\mathcal{J}$ falls.

3 With $G_t - V(s_t)$ it becomes rarer and $\mathcal{J}$ rises. One day is a small sample: on average the two updates agree (appendix B.1).

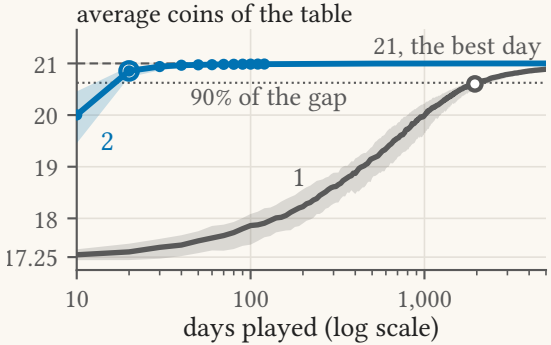
REINFORCE with a baseline converges much faster



Median of 200 runs, with the middle half. The safe step is the fastest at which 99 of 100 other runs close 90% of the gap.

1 REINFORCE at its safe step, 0.0316, closes 90% of the gap from 17.25 to 21 coins in a median of 1,950 days.

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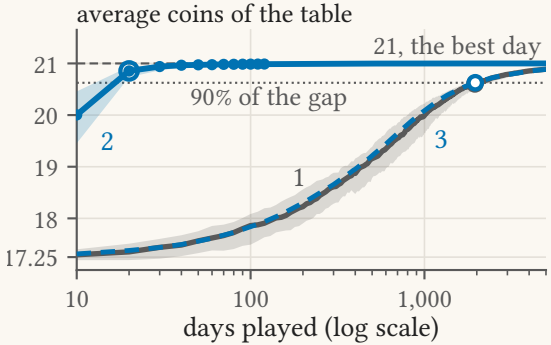


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2 With the baseline, at its safe step, 5.62, it takes 20 days: 97.5 times fewer. Here $V(s_t)$ is exact.

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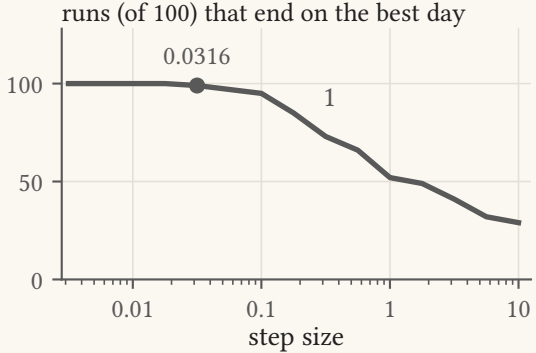
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2 With the baseline, at its safe step, 5.62, it takes 20 days: 97.5 times fewer. Here $V(s_t)$ is exact.

3 At REINFORCE's step the same baseline is no faster: 1,960 days.

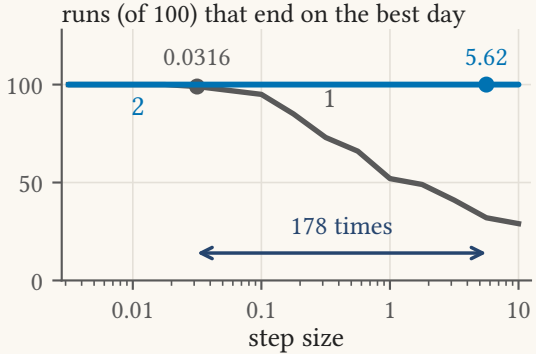
The baseline allows a much larger step



A run ends on the best day when its likelier roads lead to the 21-coin path.

1 REINFORCE fails more often above 0.0316: at 5.62, 32 of 100 runs end on the best day.

The baseline allows a much larger step

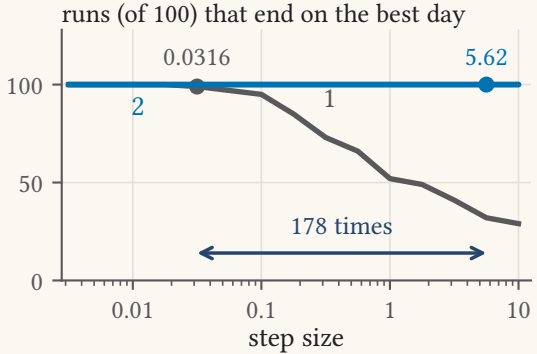


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2 With the baseline all 100 do, up to 10, the largest step tried. Its safe step is 178 times larger.

The baseline allows a much larger step



A run ends on the best day when its likelier roads lead to the 21-coin path.

1 REINFORCE fails more often above 0.0316: at 5.62, 32 of 100 runs end on the best day.

2 With the baseline all 100 do, up to 10, the largest step tried. Its safe step is 178 times larger.

3 At one common step the two take equally long, since their average push is the same. The baseline's push is quieter (appendix B), and that allows the larger step.

Here $V(s_t)$ is computed from the road itself

```
w = knobs, all 0
repeat:
  play N days with softmax(w)
1 compute V(s) for every place
```

1 $V(s)$ of every place, from the table and the road: the tree is small, so it is exact.

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w = knobs, all 0
repeat:
  play N days with softmax(w)
1 compute V(s) for every place
  push every road of every day
2   as hard as the coins after it
    minus V(s_t)
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1 compute V(s) for every place
  push every road of every day
2   as hard as the coins after it
    minus V(s_t)
3 w = w + alpha * average push
```

1 $V(s)$ of every place, from the table and the road: the tree is small, so it is exact.

2 Every road is pushed by $G_t - V(s_t)$ instead of G_t.

3 The knobs move as before. Every run of the last two frames used this exact V.

5

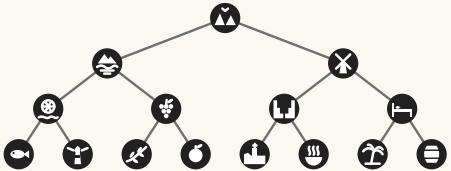
Part 5

Actor-critic

Adds. A critic: a table of numbers trained on the days to estimate $V(s_t)$, so that the multiplier $G_t - V_\phi(s_t)$ needs no list of paths.

Answers. $V(s_t)$ was an average over the eight paths of the road; on a real problem no one can list the paths.

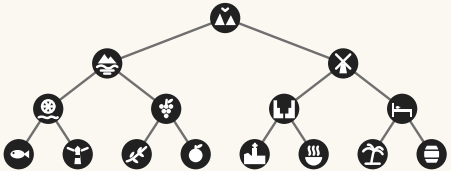
$V(s_t)$ is usually very hard to calculate



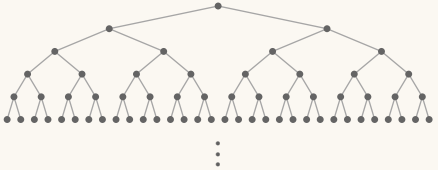
Mara's road: eight paths

1 Part 4's multiplier $\Psi_t = G_t - V(s_t)$ needs $V(s_t)$: on Mara's road an average over eight paths. We listed them.

$V(s_t)$ is usually very hard to calculate



Mara's road: eight paths

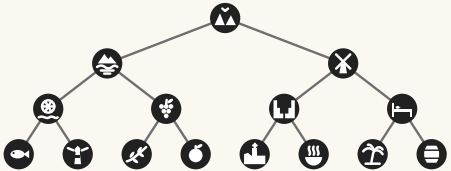


chess, a language model

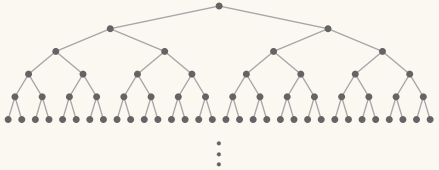
1 Part 4's multiplier $\Psi_t = G_t - V(s_t)$ needs $V(s_t)$: on Mara's road an average over eight paths. We listed them.

2 In chess, or with a language model, no one can list the paths: $V(s_t)$ is very hard to calculate.

$V(s_t)$ is usually very hard to calculate



Mara's road: eight paths



chess, a language model

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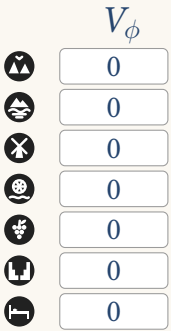
So we estimate it!

the actor
π_w, the table of knobs

→

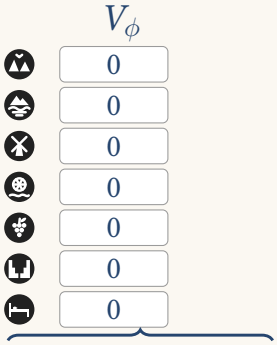
the critic
V_ϕ, an estimate of V

A critic is a function that estimates V



1 A critic is a function V_ϕ: from a place it returns a number, our estimate of V. Its weights are ϕ.

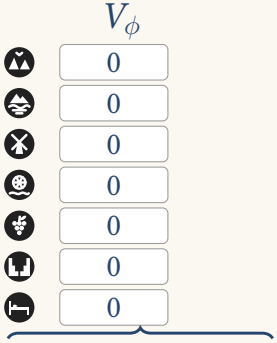
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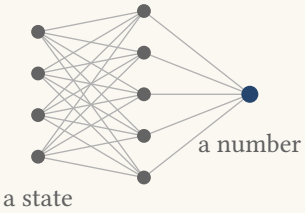
1 A critic is a function V_ϕ: from a place it returns a number, our estimate of V. Its weights are ϕ.

2 On Mara's road it is a table, one number for each place with a choice. We start it at zeros.

A critic is a function that estimates V



a day that ends is worth 0: no entry



a neural network

1 A critic is a function V_ϕ: from a place it returns a number, our estimate of V. Its weights are ϕ.

2 On Mara's road it is a table, one number for each place with a choice. We start it at zeros.

3 In real life a neural network reads the state. Either way $V_\phi(s_t)$ takes the place of $V(s_t)$ in Ψ_t.

Every step of every day is a labelled example



1 Play N days with π_w: here the five days of Part 2.

Every step of every day is a labelled example

	step 1	step 2	step 3
day 1	15	6	2
day 2	17	10	3
day 3	14	6	1
day 4	15	6	3
day 5	19	9	6

1 Play N days with π_w: here the five days of Part 2.

2 Each step is an example: the place s_t where it starts, and its label, the coins G_t from there to the end of the day.

Every step of every day is a labelled example



1 Play N days with π_w: here the five days of Part 2.

2 Each step is an example: the place s_t where it starts, and its label, the coins G_t from there to the end of the day.

3 It is a data set: learn to guess G_t from the place s_t.

The critic's loss is the squared gap

$$L(\phi) = \frac{1}{N} \sum_{n=1}^{N} \sum_{t=1}^{T} \tfrac{1}{2} \Big(\underbrace{G_t^{(n)} - V_\phi(s_t^{(n)})}_{\text{the gap}} \Big)^2$$

1 One example costs half the squared gap between the critic's number and its label.

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the labels of each place

- **15, 17, 14, 15, 19**
- **10**
- **6, 6, 6, 9**
- none
- **3**
- **2, 6**
- **1, 3**

1 One example costs half the squared gap between the critic's number and its label.

2 Each place has its own part of the loss: its labels only.

The critic's loss is the squared gap

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	the labels of each place	their average
🏔	**15, 17, 14, 15, 19**	**16.0**
🏯	**10**	**10.0**
✕	**6, 6, 6, 9**	**6.75**
🎡	none	none
🌳	**3**	**3.0**
🪑	**2, 6**	**4.0**
🛏	**1, 3**	**2.0**

1 One example costs half the squared gap between the critic's number and its label.

2 Each place has its own part of the loss: its labels only.

3 A part is smallest at the average of its labels (appendix C.1): with many days, $V(s_t)$.

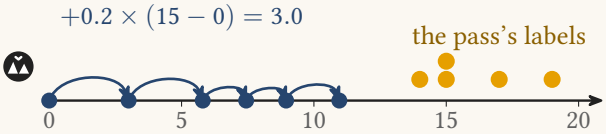
A gradient step moves the table towards the labels

$$\phi \leftarrow \phi - \beta \nabla_\phi \tfrac{1}{2}\big(G_t - V_\phi(s_t)\big)^2$$

1 Only the entry of s_t is in the loss of an example: it moves a fraction β of the way to the label. β is the critic's step.

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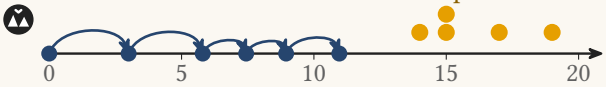


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A gradient step moves the table towards the labels

$$\phi \leftarrow \phi - \beta \nabla_\phi \tfrac{1}{2}\big(G_t - V_\phi(s_t)\big)^2$$

$+0.2 \times (15 - 0) = 3.0$



	before	after five days
	0	10.96
	0	2.00
	0	4.14
	0	0.00
	0	0.60
	0	1.52
	0	0.76

1 Only the entry of s_t is in the loss of an example: it moves a fraction β of the way to the label. β is the critic's step.

2 One step per example, in order. Here $\beta = 0.2$. A place that no day visited keeps its number.

Algorithm to estimate the critic

```
w, phi = zeros, zeros
repeat:
  play N days with softmax(w)
  G = coins to come
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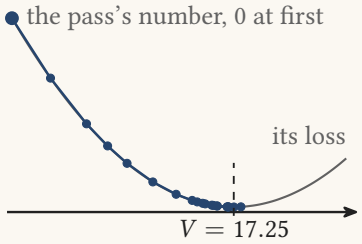
Algorithm to estimate the critic

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w, phi = zeros, zeros
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  w = w + alpha * average push
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Algorithm to estimate the critic

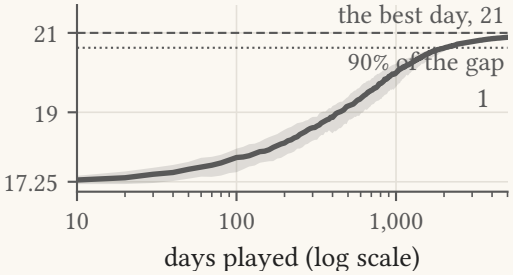
```
w, phi = zeros, zeros
repeat:
  play N days with softmax(w)
  G = coins to come
  Psi = G - phi[s]                          (1)
  w = w + alpha * average push
  for each example, in order:               (2)
    phi -= beta * grad(loss)
```



1 The multiplier $G_t - V_\phi(s_t)$ reads the table as it stood before the batch, so it never depends on the roads of the day it multiplies.

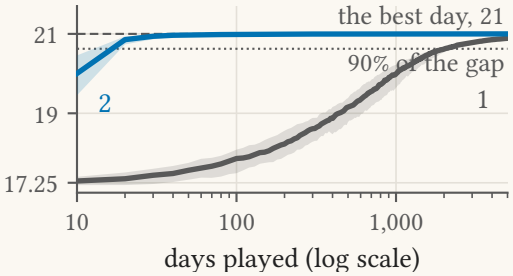
2 The critic steps down its own bowl, one example at a time.

A learned critic speeds REINFORCE up



1 REINFORCE closes 90% of the gap in 1,950 days ($\alpha = 0.0316$; median of 200 runs).

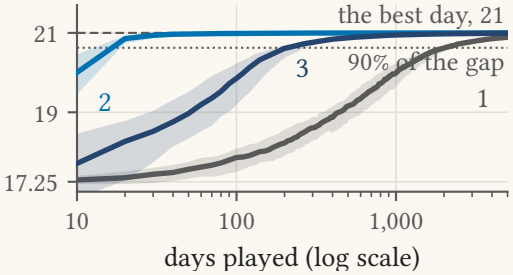
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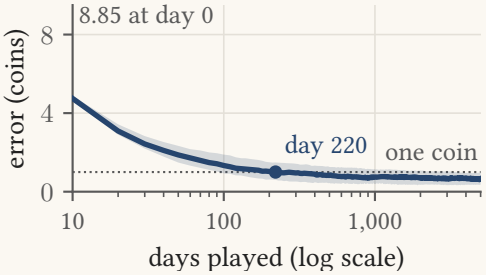


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2 With the exact V of Part 4: 20 days ($\alpha = 5.62$).

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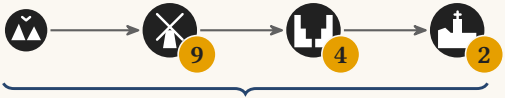


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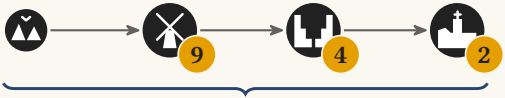
3 A critic learned from zeros: 210 days ($\alpha = 0.316$, $\beta = 0.1$).

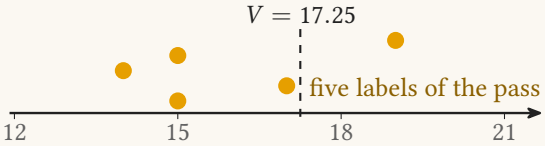
Estimating the critic is expensive



1 To label a step we wait until the day ends.

Estimating the critic is expensive

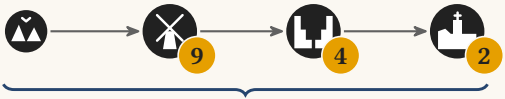


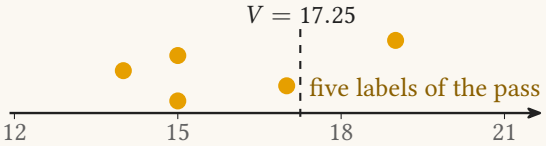


1 To label a step we wait until the day ends.

2 A label is a sum of random coins: its spread around V is 2.82 coins.

Estimating the critic is expensive





1 To label a step we wait until the day ends.

2 A label is a sum of random coins: its spread around V is 2.82 coins.

3 The critic needs many days to average that noise out. Can we find a cheaper way?

The coins to come unfold one step at a time

$$G_t = r_t + \gamma\, r_{t+1} + \gamma^2\, r_{t+2} + \cdots + \gamma^{T-t}\, r_T$$

1 G_t adds the coins from step t to the last step T of the day, each one γ times smaller than the one before. On Mara's road $T = 3$ and $\gamma = 1$.

The coins to come unfold one step at a time

$$
\begin{aligned}
G_t &= r_t + \gamma\, r_{t+1} + \gamma^2\, r_{t+2} + \cdots + \gamma^{T-t}\, r_T \\
&= r_t + \gamma \underbrace{\left(r_{t+1} + \gamma\, r_{t+2} + \cdots + \gamma^{T-t-1}\, r_T \right)}_{G_{t+1}} = r_t + \gamma\, G_{t+1}
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1 G_t adds the coins from step t to the last step T of the day, each one γ times smaller than the one before. On Mara's road $T = 3$ and $\gamma = 1$.

2 Take r_t out: what is left is γ times G_{t+1}.

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$$

$G_3 = r_3 = 2$ $\qquad$ $G_2 = r_2 + G_3 = 4 + 2 = 6$ $\qquad$ $G_1 = r_1 + G_2 = 9 + 6 = 15$

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3 Day 1 of Part 2, from the last step back.

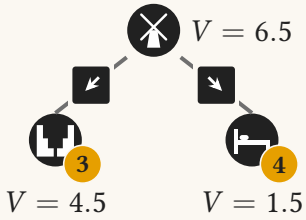
Averaging it gives the Bellman equation

$$V(s_t) = \underbrace{\mathbb{E}\big[\, r_t + \gamma\, V(s_{t+1}) \mid s_t \big]}_{\text{the average over what can follow } s_t} \;\approx\; r_t + \gamma\, V(s_{t+1})$$

1 Average both sides of $G_t = r_t + \gamma G_{t+1}$, given that Mara stands at s_t: over the road, the next place and the luck.

Averaging it gives the Bellman equation

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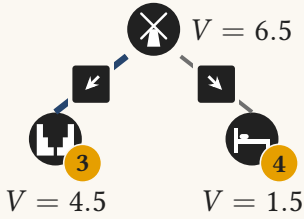


1 Average both sides of $G_t = r_t + \gamma G_{t+1}$, given that Mara stands at s_t: over the road, the next place and the luck.

2 At the market:
$\frac{1}{2}(3 + 4.5) + \frac{1}{2}(4 + 1.5) = 6.5$.

Averaging it gives the Bellman equation

$$V(s_t) = \underbrace{\mathbb{E}\big[\, r_t + \gamma\, V(s_{t+1}) \mid s_t \big]}_{\text{the average over what can follow } s_t} \;\approx\; \underbrace{r_t + \gamma\, V(s_{t+1})}_{\text{one sample of it}}$$



Day 1 took the first road: paid 4 coins, the usual 3 and a lucky one.

1 Average both sides of $G_t = r_t + \gamma G_{t+1}$, given that Mara stands at s_t: over the road, the next place and the luck.

2 At the market: $\frac{1}{2}(3 + 4.5) + \frac{1}{2}(4 + 1.5) = 6.5$.

3 One step gives one sample: $4 + 4.5 = 8.5$.

After every step the critic can learn

$$y_t = r_t + \gamma\, V_\phi(s_{t+1})$$

the target: the table's number for the next place stands in for the rest of the day

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$$\phi \;\leftarrow\; \phi - \beta\, \nabla_\phi \tfrac{1}{2}\Big(\underbrace{y_t}_{\text{held fixed}} - V_\phi(s_t)\Big)^2$$

the gradient goes through $V_\phi(s_t)$ only

1 A step of size β with y_t held fixed: a semi-gradient.

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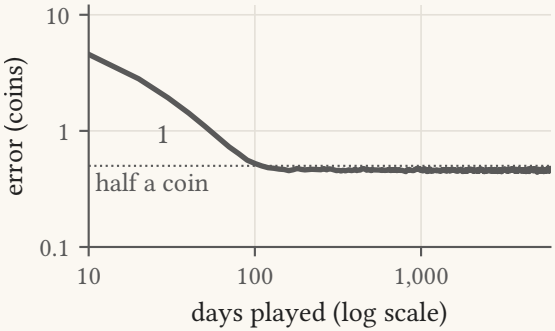
$$y = 4 + 1.52 = 5.52$$

$$V_\phi(\text{market}) = 4.14 + 0.2 \times (5.52 - 4.14) = 4.42$$

1 A step of size β with y_t held fixed: a semi-gradient.

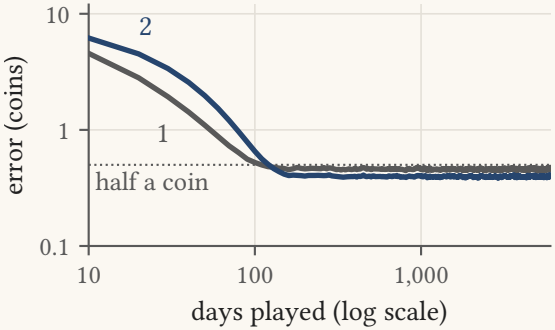
2 In the algorithm the label G_t of the critic's step becomes y_t. The multiplier is still $G_t - V_\phi(s_t)$.

A one-step label is quieter and no faster



1 Only the critic learns: the policy is fixed. Labels G_t, at the best $\beta = 0.1$: within half a coin after 110 days.

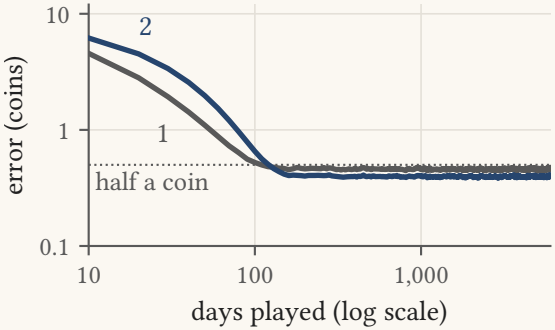
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2 One-step labels, at the best $\beta = 0.1$: 130 days.

A one-step label is quieter and no faster

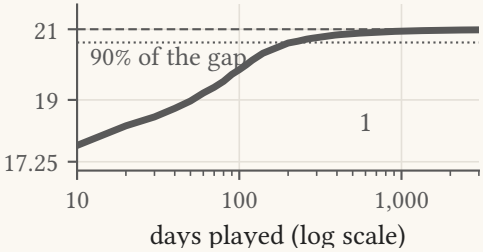


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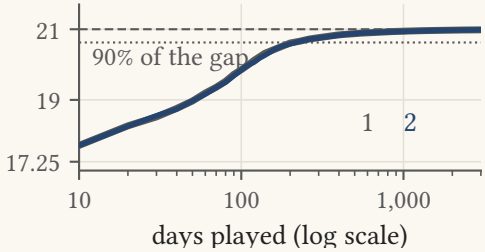
3 They are quieter (1.93 against 2.82 at the pass), but lean on a table that starts wrong.

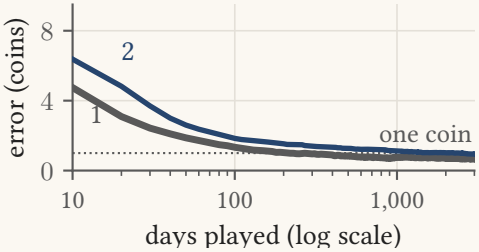
In the loop the two critics are equally fast



1 With a critic trained on returns REINFORCE needs 210 days ($\alpha = 0.316$, $\beta = 0.1$; median of 200 runs).

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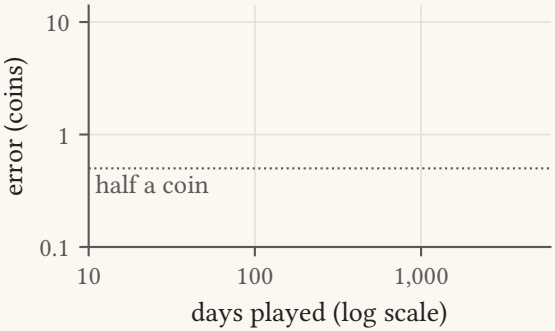




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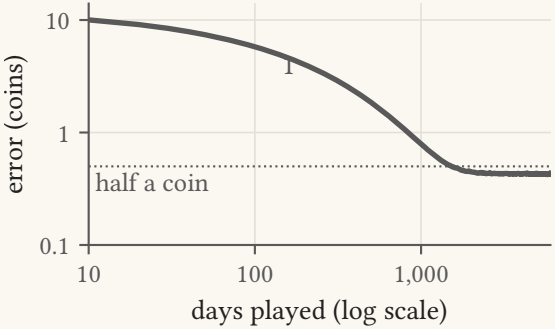
2 With one-step labels, 220 days ($\alpha = 0.316$, $\beta = 0.1$). That critic ends further from V: 0.94 coins against 0.64.

On longer roads one-step labels need fewer days



1 An extra on another road: 6 steps a day, 63 places in the table, the same luck.

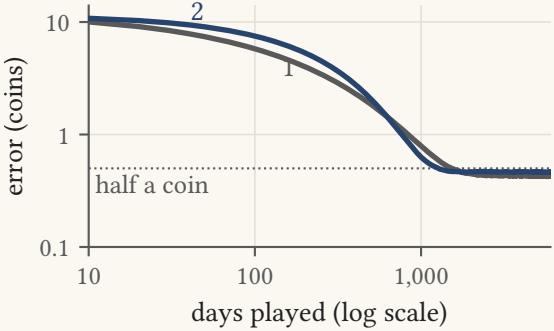
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2 Labels G_t, at the best $\beta = 0.0562$: within half a coin after 1,530 days.

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2 Labels G_t, at the best $\beta = 0.0562$: within half a coin after 1,530 days.

3 One-step labels, at $\beta = 0.1$: 1,270 days. From four steps on they need 15% to 20% fewer.

6

Part 6

GAE

Adds. A multiplier built from the critic's numbers: a surprise for every step of a day, then a sweep from the last step to the first, with one number λ.

Answers. Part 4's multiplier carries all the luck of the day. Part 5's one-step error trusts the critic for everything after the next place. One number λ chooses between them.

A third multiplier uses the critic's numbers

the box Ψ_t of the gradient formula holds

Part 3 $\Psi_t = \boxed{G_t}$

the coins to come

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Parts 4 and 5 $\Psi_t = \boxed{G_t - V(s_t)}$

the coins to come minus the critic's number

1 The box has held the coins to come, then the coins to come minus the critic's number.

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Part 6 $\Psi_t = \boxed{\hat{A}_t}$

built from the critic's numbers

1 The box has held the coins to come, then the coins to come minus the critic's number.

2 A third option, $\hat{A}_t$, starts from the critic's numbers.

A third multiplier uses the critic's numbers

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built from the critic's numbers

1 The box has held the coins to come, then the coins to come minus the critic's number.

2 A third option, $\hat{A}_t$, starts from the critic's numbers.

3 One number λ sets how much of the day it lets in.

Step 1: a surprise for every step, all at once

$$\delta_t = r_t + \gamma\, V(s_{t+1}) - V(s_t) \quad \text{for } t = 1, \ldots, T$$

1 The surprise of a step: what Mara got, plus what the critic expects next, minus what it expected here.

Step 1: a surprise for every step, all at once

$$\delta_t = r_t + \gamma\, V(s_{t+1}) - V(s_t) \quad \text{for } t = 1, \ldots, T$$

	step 1	step 2	step 3
			
r_t	9	4	2
$+\,\gamma V(s_{t+1})$	6.5	4.5	0
$-\,V(s_t)$	17.25	6.5	4.5

1 The surprise of a step: what Mara got, plus what the critic expects next, minus what it expected here.

2 V is the critic's number for a place; the end is worth 0. Here $T = 3$ and $\gamma = 1$.

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$=\ \delta_t$	−1.75	+2	−2.5

1 The surprise of a step: what Mara got, plus what the critic expects next, minus what it expected here.

2 V is the critic's number for a place; the end is worth 0. Here $T = 3$ and $\gamma = 1$.

3 Positive: the step beat what the critic expected. Negative: it fell short.

Step 2 sweeps from the last step to the first

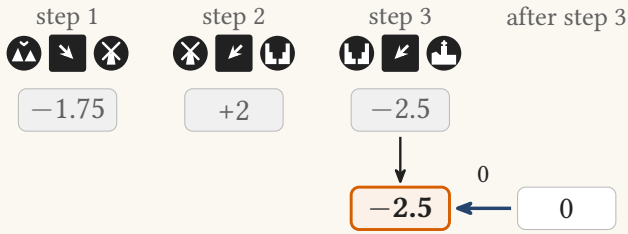
$$\hat{A}_{T+1} = 0$$
$$\hat{A}_t = \delta_t + \gamma\lambda\,\hat{A}_{t+1} \qquad \text{for } t = T, T-1, \ldots, 1$$

	step 1	step 2	step 3	after step 3
				
δ_t	-1.75	$+2$	-2.5	
$\hat{A}_t$				0

1 The sweep starts after the last step, with $\hat{A}_{T+1} = 0$.

Step 2 sweeps from the last step to the first

$$\hat{A}_{T+1} = 0$$
$$\hat{A}_t = \delta_t + \gamma\lambda\,\hat{A}_{t+1} \qquad \text{for } t = T, T-1, \ldots, 1$$

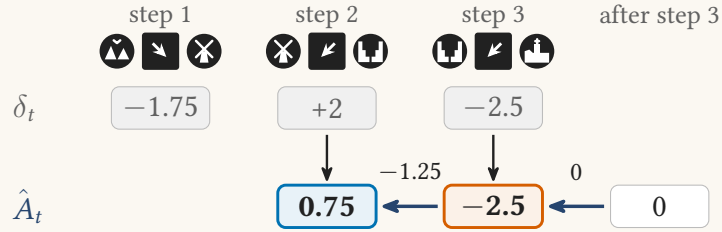


1 The sweep starts after the last step, with $\hat{A}_{T+1} = 0$.

2 Each step adds its surprise to $\gamma\lambda$ times the value of the step after it. Here $\gamma = 1$ and $\lambda = 0.5$.

Step 2 sweeps from the last step to the first

$$\hat{A}_{T+1} = 0$$
$$\hat{A}_t = \delta_t + \gamma\lambda\,\hat{A}_{t+1} \qquad \text{for } t = T, T-1, \ldots, 1$$

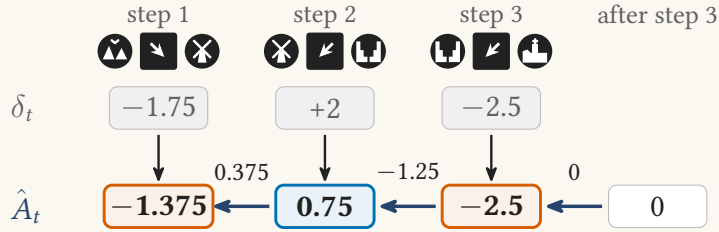


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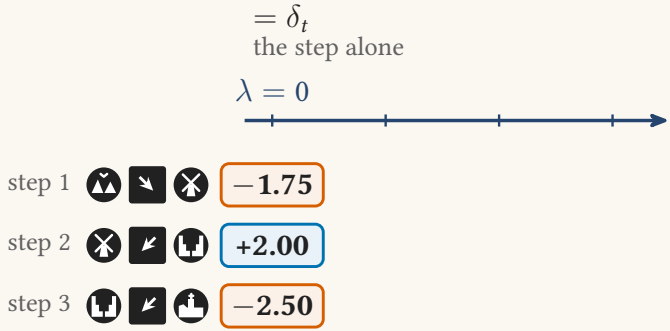


1 The sweep starts after the last step, with $\hat{A}_{T+1} = 0$.

2 Each step adds its surprise to $\gamma\lambda$ times the value of the step after it. Here $\gamma = 1$ and $\lambda = 0.5$.

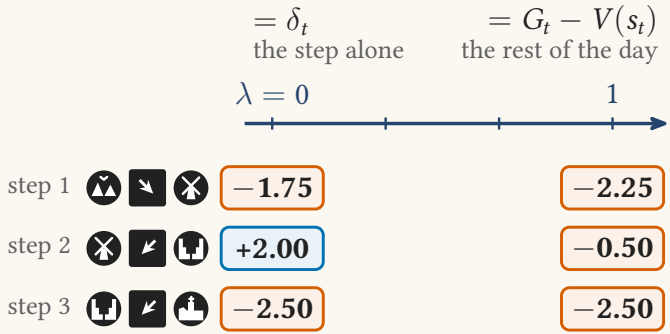
3 It runs from the last step to the first.

One number λ dials between two multipliers



1 The multiplier is $\Psi_t = \hat{A}_t$, the value the sweep ends with at step t.

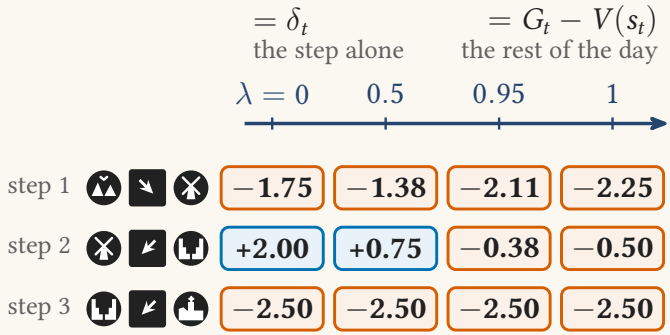
One number λ dials between two multipliers



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2 $\lambda = 0$ gives δ_t. $\lambda = 1$ gives $G_t - V(s_t)$, the multiplier of Part 4: $15 - 17.25 = -2.25$ at step 1.

One number λ dials between two multipliers

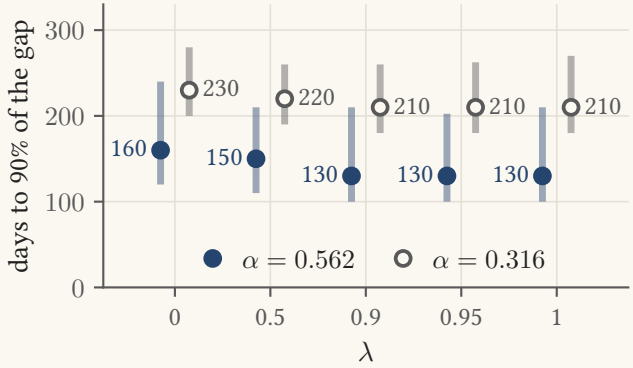


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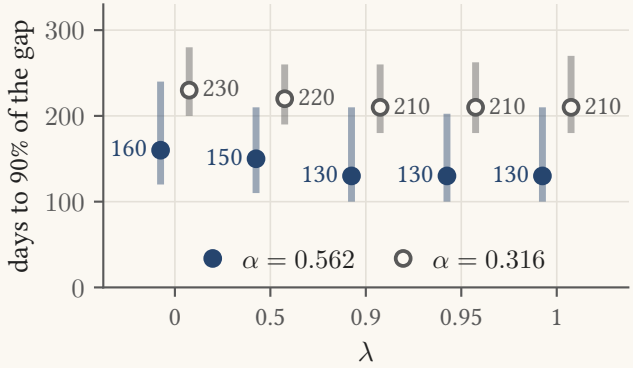
3 In between, λ sets how much of the rest of the day is let in.

On a three-step road the dial matters little



1 Median (dot) and middle half (bar) of 200 runs, until 90% of the gap from 17.25 to 21 coins is closed.

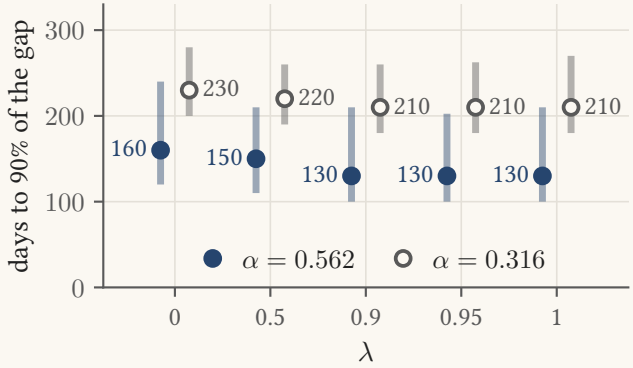
On a three-step road the dial matters little



1 Median (dot) and middle half (bar) of 200 runs, until 90% of the gap from 17.25 to 21 coins is closed.

2 At $\alpha = 0.562$ the runs are faster, but 6 to 9 in 200 never get there. At $\alpha = 0.316$ none fails.

On a three-step road the dial matters little



1 Median (dot) and middle half (bar) of 200 runs, until 90% of the gap from 17.25 to 21 coins is closed.

2 At $\alpha = 0.562$ the runs are faster, but 6 to 9 in 200 never get there. At $\alpha = 0.316$ none fails.

3 For scale, REINFORCE takes 1,950 days.

A library writes the sweep as a short loop

```
delta = coins + gamma * v_next - v
```

1 One line gives every surprise: v and v_next hold $V(s_t)$ and $V(s_{t+1})$.

A library writes the sweep as a short loop

```
delta = coins + gamma * v_next - v
run = torch.zeros(len(delta))
adv = torch.zeros_like(delta)
for t in reversed(range(3)):
```

1 One line gives every surprise: `v` and `v_next` hold $V(s_t)$ and $V(s_{t+1})$.

2 `run` starts at 0, and the loop goes from the last step to the first.

A library writes the sweep as a short loop

```
delta = coins + gamma * v_next - v
run = torch.zeros(len(delta))
adv = torch.zeros_like(delta)
for t in reversed(range(3)):
    run = delta[:, t] + gamma * lam * run
    adv[:, t] = run
```

1 One line gives every surprise: `v` and `v_next` hold $V(s_t)$ and $V(s_{t+1})$.

2 `run` starts at 0, and the loop goes from the last step to the first.

3 `adv` has the shape of G in Part 3: it is the multiplier.

7

Part 7

PPO

Adds. PPO is REINFORCE with a new loss. The loss remembers the table that played the days, through the ratio of the new chance of each road to its old chance.

Answers. Part 3's loss forgets which table played the days.

REINFORCE is gradient descent on a loss

$$L(w) = -\frac{1}{N}\sum_{n=1}^{N}\sum_{t=1}^{T} \boxed{\Psi_t^{(n)}} \log \pi_w(a_t^{(n)} \mid s_t^{(n)})$$

1 This is Part 3's loss: minus the multiplier times the log chance of the road taken.

REINFORCE is gradient descent on a loss

$$L(w) = -\frac{1}{N}\sum_{n=1}^{N}\sum_{t=1}^{T} \boxed{\Psi_t^{(n)}} \log \pi_w(a_t^{(n)} \mid s_t^{(n)})$$

$$\nabla L(w) = -\frac{1}{N}\sum_{n=1}^{N}\sum_{t=1}^{T} \underbrace{\boxed{\Psi_t^{(n)}} \nabla_w \log \pi_w(a_t^{(n)} \mid s_t^{(n)})}_{\text{minus the estimate of Part 3}}$$

1. This is Part 3's loss: minus the multiplier times the log chance of the road taken.
2. The multiplier is a plain number: the gradient is minus the estimate of Part 3.

REINFORCE is gradient descent on a loss

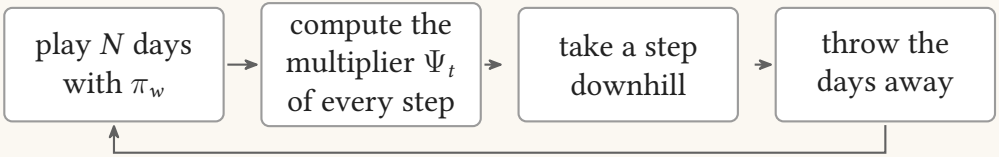
$$L(w) \;=\; -\frac{1}{N}\sum_{n=1}^{N}\sum_{t=1}^{T} \boxed{\Psi_t^{(n)}} \quad \log \pi_w(a_t^{(n)} \mid s_t^{(n)})$$

$$\nabla L(w) \;=\; -\frac{1}{N}\sum_{n=1}^{N}\sum_{t=1}^{T} \boxed{\Psi_t^{(n)}} \quad \nabla_w \log \pi_w(a_t^{(n)} \mid s_t^{(n)})$$

minus the estimate of Part 3

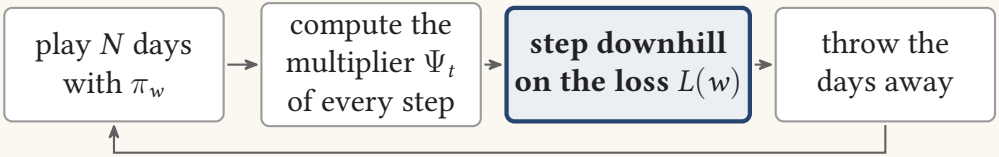
1. This is Part 3's loss: minus the multiplier times the log chance of the road taken.
2. The multiplier is a plain number: the gradient is minus the estimate of Part 3.
3. A step downhill, $w \leftarrow w - \alpha\,\nabla L(w)$, is a step uphill on $\mathcal{J}$: Part 3's update.

PPO keeps the algorithm and changes the loss



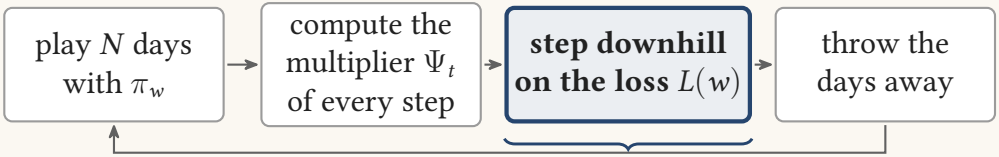
1 The algorithm of Part 3 stays: play, compute the multiplier, step, discard.

PPO keeps the algorithm and changes the loss



1. The algorithm of Part 3 stays: play, compute the multiplier, step, discard.
2. The step is gradient descent on a loss: the loss decides what the step does.

PPO keeps the algorithm and changes the loss



Part 3 $L(w) = -\frac{1}{N} \sum \sum \Psi_t \log \pi_w(a_t \mid s_t)$

Part 7 a new loss, built in the next frames

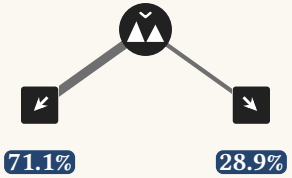
1 The algorithm of Part 3 stays: play, compute the multiplier, step, discard.

2 The step is gradient descent on a loss: the loss decides what the step does.

3 Autograd differentiates whatever loss we write. PPO writes a new one.

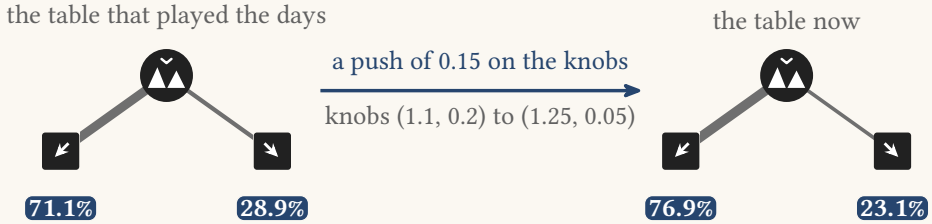
The loss remembers the table that played the days

the table that played the days



1 Each step stores the chance its road had in the table that played the day: $\pi_{\text{old}}(a_t \mid s_t)$.

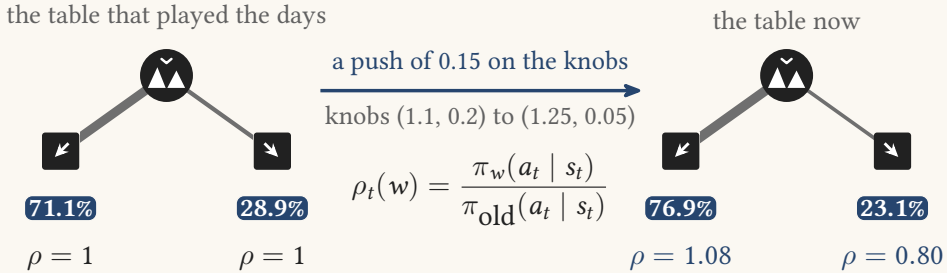
The loss remembers the table that played the days



1 Each step stores the chance its road had in the table that played the day: $\pi_{\text{old}}(a_t \mid s_t)$.

2 After a push the table gives other chances: 76.9% and 23.1% at the pass.

The loss remembers the table that played the days



1 Each step stores the chance its road had in the table that played the day: $\pi_{\text{old}}(a_t \mid s_t)$.

2 After a push the table gives other chances: 76.9% and 23.1% at the pass.

3 The ratio is the new chance over the old one: 1.08 for ↙ and 0.80 for ↘.

The first loss multiplies the ratio by the multiplier

$$L(w) = -\frac{1}{N}\sum_{n=1}^{N}\sum_{t=1}^{T} \boxed{\Psi_t^{(n)}} \log \pi_w(a_t^{(n)} \mid s_t^{(n)})$$

1 Part 3's loss has the log chance of the road taken in every term.

The first loss multiplies the ratio by the multiplier

$$L(w) = -\frac{1}{N}\sum_{n=1}^{N}\sum_{t=1}^{T} \boxed{\Psi_t^{(n)}} \quad \log \pi_w(a_t^{(n)} \mid s_t^{(n)})$$

replace it by the ratio

$$L(w) = -\frac{1}{N}\sum_{n=1}^{N}\sum_{t=1}^{T} \boxed{\Psi_t^{(n)}} \quad \rho_t^{(n)}(w)$$

1 Part 3's loss has the log chance of the road taken in every term.

2 The new loss uses the ratio instead. At the old table every ratio is 1.

The first loss multiplies the ratio by the multiplier

$$L(w) = -\frac{1}{N}\sum_{n=1}^{N}\sum_{t=1}^{T} \boxed{\Psi_t^{(n)}} \quad \log \pi_w(a_t^{(n)} \mid s_t^{(n)})$$

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1 Part 3's loss has the log chance of the road taken in every term.

2 The new loss uses the ratio instead. At the old table every ratio is 1.

3 So the gradient is Part 3's and the first step is Part 3's step (appendix D.1).

Appendix

The algebra behind Part 4

Nothing in the frames that follow is needed to follow the part.

B.1 a baseline of the place leaves the average push unchanged
B.2 a baseline shrinks the spread of the push

Appendix B.1: a baseline leaves the push alone

$V(s_t)$ is fixed when Mara stands at s_t, before road a_t. score_t is short for $\nabla_w \log \pi_w(a_t \mid s_t)$; given s_t its mean is zero, since $\sum_a \nabla \pi(a \mid s) = \nabla 1 = 0$. The luck averages out.

$$\mathbb{E}\big[(G_t - V(s_t))\,\text{score}_t\big] = \mathbb{E}\big[G_t\,\text{score}_t\big] - \mathbb{E}\big[V(s_t)\,\text{score}_t\big]$$

difference of averages

$$\mathbb{E}\big[V(s_t)\,\text{score}_t\big]$$

$$\mathbb{E}\big[(G_t - V(s_t))\,\text{score}_t\big]$$

Appendix B.1: a baseline leaves the push alone

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difference of averages

$$\mathbb{E}\big[V(s_t)\,\text{score}_t\big] = \mathbb{E}\Big[\,V(s_t)\;\mathbb{E}\big[\text{score}_t \mid s_t\big]\Big]$$

$V(s_t)$ is fixed given s_t

$$\mathbb{E}\big[(G_t - V(s_t))\,\text{score}_t\big]$$

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$$\mathbb{E}\big[V(s_t)\,\text{score}_t\big] = \mathbb{E}\Big[\,V(s_t)\;\mathbb{E}\big[\text{score}_t \mid s_t\big]\Big]$$

$V(s_t)$ is fixed given s_t

$$= \mathbb{E}\big[\,V(s_t)\cdot 0\,\big] \;=\; 0$$

the score has mean zero

$$\mathbb{E}\big[(G_t - V(s_t))\,\text{score}_t\big]$$

Appendix B.1: a baseline leaves the push alone

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$$\mathbb{E}\big[(G_t - V(s_t))\,\text{score}_t\big] = \mathbb{E}\big[G_t\,\text{score}_t\big] - \mathbb{E}\big[V(s_t)\,\text{score}_t\big] \qquad \textit{difference of averages}$$

$$\mathbb{E}\big[V(s_t)\,\text{score}_t\big] = \mathbb{E}\Big[\,V(s_t)\;\mathbb{E}\big[\text{score}_t \mid s_t\big]\Big] \qquad V(s_t)\ \textit{is fixed given}\ s_t$$

$$= \mathbb{E}\big[\,V(s_t)\cdot 0\,\big] \;=\; 0 \qquad \textit{the score has mean zero}$$

$$\mathbb{E}\big[(G_t - V(s_t))\,\text{score}_t\big] = \mathbb{E}\big[G_t\,\text{score}_t\big] \qquad \textit{lines 1 and 3}$$

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$$\mathbb{E}\big[(G_t - V(s_t))\,\text{score}_t\big] = \mathbb{E}\big[G_t\,\text{score}_t\big] - \mathbb{E}\big[V(s_t)\,\text{score}_t\big] \qquad \textit{difference of averages}$$

$$\mathbb{E}\big[V(s_t)\,\text{score}_t\big] = \mathbb{E}\Big[\,V(s_t)\;\mathbb{E}\big[\text{score}_t \mid s_t\big]\Big] \qquad V(s_t)\ \textit{is fixed given}\ s_t$$

$$= \mathbb{E}\big[\,V(s_t)\cdot 0\,\big] \;=\; 0 \qquad \textit{the score has mean zero}$$

$$\mathbb{E}\big[(G_t - V(s_t))\,\text{score}_t\big] = \mathbb{E}\big[G_t\,\text{score}_t\big] \qquad \textit{lines 1 and 3}$$

Check on the road. 6 tables of knobs, 9 baselines of the place each: $|\mathbb{E}[b(s_t)\,\text{score}_t]| \le 7 \cdot 10^{-14}$ and the average push of $G_t - b(s_t)$ is the exact gradient of $\mathcal{J}$ to $2 \cdot 10^{-14}$ (check_derivations.py, D1).

Appendix B.2: a baseline shrinks the spread

One day at the uniform table: its push on the first-road knob of the pass is $\Psi_1\, s$, with $s = \mathbf{1}[a_1 = \blacktriangleleft] - \frac{1}{2}$. Var $G_1 = 7.94$: 5.94 between the paths and 2 from the luck.

$$\mathbb{E}\big[\Psi_1\, s\big] = 0.875 \text{ for both multipliers}$$

B.1; the exact slope of $\mathcal{J}$

$$\mathbb{E}\big[(G_1\, s)^2\big]$$

$$\mathbb{E}\big[((G_1 - V(s_1))\, s)^2\big]$$

spread

Appendix B.2: a baseline shrinks the spread

One day at the uniform table: its push on the first-road knob of the pass is $\Psi_1\, s$, with $s = \mathbf{1}[a_1 = \blacktriangleleft] - \frac{1}{2}$. Var $G_1 = 7.94$: 5.94 between the paths and 2 from the luck.

$$\mathbb{E}\big[\Psi_1\, s\big] = 0.875 \text{ for both multipliers}$$

$$\mathbb{E}\big[(G_1\, s)^2\big] = \tfrac{1}{4}\big(V(s_1)^2 + \operatorname{Var} G_1\big) = 76.4$$

$$\mathbb{E}\big[((G_1 - V(s_1))\, s)^2\big]$$

spread

B.1; the exact slope of $\mathfrak{J}$

$s^2 = \frac{1}{4}$; $\mathbb{E}[G^2] = V^2 + \operatorname{Var} G$

Appendix B.2: a baseline shrinks the spread

One day at the uniform table: its push on the first-road knob of the pass is $\Psi_1\, s$, with $s = \mathbf{1}[a_1 = \blacktriangleleft] - \frac{1}{2}$. Var $G_1 = 7.94$: 5.94 between the paths and 2 from the luck.

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$s^2 = \frac{1}{4}$; $\mathbb{E}[G^2] = V^2 + \operatorname{Var} G$

the mean taken out

Appendix B.2: a baseline shrinks the spread

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$$\text{spread} = \sqrt{\mathbb{E}[(\Psi_1\, s)^2] - 0.875^2} \;:\; 8.70,\; 1.10$$

B.1; the exact slope of $\mathfrak{J}$

$s^2 = \frac{1}{4}$; $\mathbb{E}[G^2] = V^2 + \operatorname{Var} G$

the mean taken out

variance formula

Appendix B.2: a baseline shrinks the spread

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B.1; the exact slope of $\mathcal{J}$

$s^2 = \frac{1}{4}$; $\mathbb{E}[G^2] = V^2 + \operatorname{Var} G$

the mean taken out

variance formula

A batch of 10 days: the spread of the average push is 2.75 with G_t and 0.35 with the baseline, around 0.875.

Appendix B.2: a baseline shrinks the spread

One day at the uniform table: its push on the first-road knob of the pass is $\Psi_1\, s$, with $s = \mathbf{1}[a_1 = \blacktriangleleft] - \frac{1}{2}$. Var $G_1 = 7.94$: 5.94 between the paths and 2 from the luck.

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$s^2 = \frac{1}{4}$; $\mathbb{E}[G^2] = V^2 + \operatorname{Var} G$

the mean taken out

variance formula

A batch of 10 days: the spread of the average push is 2.75 with G_t and 0.35 with the baseline, around 0.875.

Check on the road. All 216 cases of a day give these numbers exactly; 2,000,000 sampled days give the same spreads to within 1% (check_derivations.py, D2).

Appendix

The algebra behind Part 5

Nothing in the frames that follow is needed to follow the part.

C.1 the loss is smallest at the average
C.2 the Bellman equation
C.3 the target as a multiplier

Appendix C.1: the loss is smallest at the average

Place s has k examples with labels $y_1, \dots, y_k$; ϕ_s is its entry and L_s its part of the loss.

$$L_s(v) = \tfrac{1}{N} \textstyle\sum_i \tfrac{1}{2} (y_i - v)^2$$

only the examples of s hold ϕ_s

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$$L_s(v) = \tfrac{1}{N} \textstyle\sum_i \tfrac{1}{2} (y_i - v)^2$$

$$L_s'(v) = -\tfrac{1}{N} \textstyle\sum_i (y_i - v) = 0 \iff v = \tfrac{1}{k} \textstyle\sum_i y_i$$

only the examples of s hold ϕ_s

$L_s'' = k/N > 0$*: a minimum*

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$$\nabla_\phi \tfrac{1}{2} \big(y - \phi_s\big)^2 = -(y - \phi_s)\, \mathbf{e}_s$$

only the examples of s hold ϕ_s

$L_s'' = k/N > 0$: a minimum

one example, one entry

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Place s has k examples with labels $y_1, \dots, y_k$; ϕ_s is its entry and L_s its part of the loss.

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So the gradient step of an example is $\phi_s \leftarrow \phi_s + \beta\,(y - \phi_s)$. With many days the average of the labels of s is $V(s)$.

Appendix C.1: the loss is smallest at the average

Place s has k examples with labels $y_1, \dots, y_k$; ϕ_s is its entry and L_s its part of the loss.

$$L_s(v) = \tfrac{1}{N} \textstyle\sum_i \tfrac{1}{2} (y_i - v)^2$$

$$L_s'(v) = -\tfrac{1}{N} \textstyle\sum_i (y_i - v) = 0 \iff v = \tfrac{1}{k} \textstyle\sum_i y_i$$

$$\nabla_\phi \tfrac{1}{2} \big(y - \phi_s\big)^2 = -(y - \phi_s)\, \mathbf{e}_s$$

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one example, one entry

So the gradient step of an example is $\phi_s \leftarrow \phi_s + \beta\,(y - \phi_s)$. With many days the average of the labels of s is $V(s)$.

Check on the road. Five days and 50 seeded batches: the label averages beat 200 nearby tables each time, with gradient at most $2 \cdot 10^{-15}$. An example's gradient is $-(y - \phi_s)\,\mathbf{e}_s$ (autograd, $1 \cdot 10^{-16}$). On 14 tables the average of G_t over the 216 days through a place equals V to $1 \cdot 10^{-14}$.

Appendix C.2: the Bellman equation

A softmax table, $\gamma = 1$. Road a at s pays $r(s, a)$, the usual coins of the place $s'(s, a)$ it leads to. The luck has mean 0.

$$V(s) = \mathbb{E}\big[G_t \mid s_t = s\big] = \mathbb{E}\big[r_t + G_{t+1} \mid s_t = s\big]$$

$$G_t = r_t + G_{t+1}$$

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$$V(s) = \mathbb{E}\big[G_t \mid s_t = s\big] = \mathbb{E}\big[r_t + G_{t+1} \mid s_t = s\big]$$
$$= \sum_a \pi_w(a \mid s)\Big(r(s,a) + \mathbb{E}\big[G_{t+1} \mid s_t = s,\ a_t = a\big]\Big)$$

$G_t = r_t + G_{t+1}$

average over the road first

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&= \textstyle\sum_a \pi_w(a \mid s)\Big(r(s,a) + V\big(s'(s,a)\big)\Big)
\end{aligned}
$$

$G_t = r_t + G_{t+1}$

average over the road first

the rest of the day restarts at s'

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For a table $V_\phi = V + e$ the one-step label averages to $V(s) + \mathbb{E}[e(s_{t+1}) \mid s_t = s]$: right as far as the next number is.

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$G_t = r_t + G_{t+1}$

average over the road first

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For a table $V_\phi = V + e$ the one-step label averages to $V(s) + \mathbb{E}[e(s_{t+1}) \mid s_t = s]$: right as far as the next number is.

Check on the road. Seven tables on the road with luck, seven on a depth-6 tree: V from its definition equals the right-hand side to $1 \cdot 10^{-14}$; $G_t = r_t + G_{t+1}$ on every outcome ($1 \cdot 10^{-16}$); the label of a wrong table averages to $V(s)$ plus the next error to $2 \cdot 10^{-14}$.

Appendix C.3: the target as a multiplier

With the exact V: $\delta_t = r_t + V(s_{t+1}) - V(s_t)$, the target minus the number of the place. $\text{score}_t = \nabla_w \log \pi_w(a_t \mid s_t)$; $A(s,a) = \mathbb{E}[G_t \mid s_t = s,\ a_t = a] - V(s)$.

$$\mathbb{E}\big[\delta_t \mid s_t, a_t\big] = r(s_t, a_t) + V(s_{t+1}) - V(s_t) = A(s_t, a_t)$$

C.2 for one road

Appendix C.3: the target as a multiplier

With the exact V: $\delta_t = r_t + V(s_{t+1}) - V(s_t)$, the target minus the number of the place. $\text{score}_t = \nabla_w \log \pi_w(a_t \mid s_t)$; $A(s,a) = \mathbb{E}[G_t \mid s_t = s,\ a_t = a] - V(s)$.

$$\mathbb{E}\big[\delta_t \mid s_t, a_t\big] = r(s_t, a_t) + V(s_{t+1}) - V(s_t) = A(s_t, a_t)$$

$$\mathbb{E}\big[\delta_t \,\text{score}_t\big] = \mathbb{E}\big[(G_t - V(s_t))\,\text{score}_t\big]$$

C.2 for one road

both average to A given s_t, a_t

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$$\mathbb{E}\big[\delta_t \,\text{score}_t\big] = \mathbb{E}\big[(G_t - V(s_t))\,\text{score}_t\big]$$

both average to A given s_t, a_t

$$\mathbb{E}\Big[\sum_t \delta_t \,\text{score}_t\Big] = \nabla_w \mathcal{J}(w)$$

Part 4: a number of the place adds 0

Appendix C.3: the target as a multiplier

With the exact V: $\delta_t = r_t + V(s_{t+1}) - V(s_t)$, the target minus the number of the place. $\text{score}_t = \nabla_w \log \pi_w(a_t \mid s_t)$; $A(s,a) = \mathbb{E}[G_t \mid s_t = s,\ a_t = a] - V(s)$.

$$\mathbb{E}\big[\delta_t \mid s_t, a_t\big] = r(s_t, a_t) + V(s_{t+1}) - V(s_t) = A(s_t, a_t)$$

C.2 for one road

$$\mathbb{E}\big[\delta_t \, \text{score}_t\big] = \mathbb{E}\big[(G_t - V(s_t)) \, \text{score}_t\big]$$

both average to A given s_t, a_t

$$\mathbb{E}\Big[\sum_t \delta_t \, \text{score}_t\Big] = \nabla_w \mathcal{J}(w)$$

Part 4: a number of the place adds 0

On a day δ_t and $G_t - V(s_t)$ differ, by the luck and the roads still to come; their averages agree. This is the next part.

Appendix C.3: the target as a multiplier

With the exact V: $\delta_t = r_t + V(s_{t+1}) - V(s_t)$, the target minus the number of the place. $\text{score}_t = \nabla_w \log \pi_w(a_t \mid s_t)$; $A(s,a) = \mathbb{E}[G_t \mid s_t = s,\ a_t = a] - V(s)$.

$$\mathbb{E}\big[\delta_t \mid s_t, a_t\big] = r(s_t, a_t) + V(s_{t+1}) - V(s_t) = A(s_t, a_t)$$

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$$\mathbb{E}\big[\delta_t \,\text{score}_t\big] = \mathbb{E}\big[(G_t - V(s_t))\,\text{score}_t\big]$$

both average to A given s_t, a_t

$$\mathbb{E}\Big[\sum_t \delta_t \,\text{score}_t\Big] = \nabla_w \mathcal{J}(w)$$

Part 4: a number of the place adds 0

On a day δ_t and $G_t - V(s_t)$ differ, by the luck and the roads still to come; their averages agree. This is the next part.

Check on the road. Fourteen tables: δ_t averages to the gap to $7 \cdot 10^{-15}$. The average push equals the exact gradient to $4 \cdot 10^{-15}$ and a finite-difference one to $3 \cdot 10^{-9}$.

Appendix

The algebra behind PPO

Nothing in the frames that follow is needed to follow Part 7.

D.1 the gradient of the ratio term, and the first step
D.2 the ratio reweights the days of the old table

Appendix D.1: the gradient of the ratio term

π_w and π_{old} stand for $\pi_w(a_t \mid s_t)$ and $\pi_{\text{old}}(a_t \mid s_t)$, and $\rho_t = \pi_w / \pi_{\text{old}}$. The stored chance π_{old} is a number, and so is Ψ_t.

$$\nabla_w \rho_t = \nabla_w \pi_w \,/\, \pi_{\text{old}}$$

the denominator is a constant

Appendix D.1: the gradient of the ratio term

π_w and π_{old} stand for $\pi_w(a_t \mid s_t)$ and $\pi_{\text{old}}(a_t \mid s_t)$, and $\rho_t = \pi_w / \pi_{\text{old}}$. The stored chance π_{old} is a number, and so is Ψ_t.

$$\begin{aligned} \nabla_w \rho_t &= \nabla_w \pi_w \,/\, \pi_{\text{old}} \\ &= \pi_w \, \nabla_w \log \pi_w \,/\, \pi_{\text{old}} \end{aligned}$$

the denominator is a constant

$\nabla \pi = \pi \, \nabla \log \pi$, *the chain rule*

Appendix D.1: the gradient of the ratio term

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$$
\begin{aligned}
\nabla_w \rho_t &= \nabla_w \pi_w \,/\, \pi_{\text{old}} && \textit{the denominator is a constant} \\
&= \pi_w \nabla_w \log \pi_w \,/\, \pi_{\text{old}} && \nabla \pi = \pi \nabla \log \pi\textit{, the chain rule} \\
&= \rho_t \nabla_w \log \pi_w && \textit{the definition of the ratio}
\end{aligned}
$$

Appendix D.1: the gradient of the ratio term

π_w and π_{old} stand for $\pi_w(a_t \mid s_t)$ and $\pi_{\text{old}}(a_t \mid s_t)$, and $\rho_t = \pi_w / \pi_{\text{old}}$. The stored chance π_{old} is a number, and so is Ψ_t.

$$
\begin{aligned}
\nabla_w \rho_t &= \nabla_w \pi_w \,/\, \pi_{\text{old}} && \textit{the denominator is a constant} \\
&= \pi_w \nabla_w \log \pi_w \,/\, \pi_{\text{old}} && \nabla \pi = \pi \nabla \log \pi\textit{, the chain rule} \\
&= \rho_t \nabla_w \log \pi_w && \textit{the definition of the ratio} \\
\nabla_w (\rho_t \Psi_t) &= \rho_t \Psi_t \nabla_w \log \pi_w && \Psi_t \textit{ is a constant}
\end{aligned}
$$

Appendix D.1: the gradient of the ratio term

π_w and π_{old} stand for $\pi_w(a_t \mid s_t)$ and $\pi_{\text{old}}(a_t \mid s_t)$, and $\rho_t = \pi_w / \pi_{\text{old}}$. The stored chance π_{old} is a number, and so is Ψ_t.

$$
\begin{aligned}
\nabla_w \rho_t &= \nabla_w \pi_w \,/\, \pi_{\text{old}} && \text{the denominator is a constant} \\
&= \pi_w \, \nabla_w \log \pi_w \,/\, \pi_{\text{old}} && \nabla \pi = \pi \, \nabla \log \pi, \text{ the chain rule} \\
&= \rho_t \, \nabla_w \log \pi_w && \text{the definition of the ratio} \\
\nabla_w (\rho_t \Psi_t) &= \rho_t \, \Psi_t \, \nabla_w \log \pi_w && \Psi_t \text{ is a constant} \\
&= \Psi_t \, \nabla_w \log \pi_w \quad \text{at } w = w_{\text{old}} && \text{every ratio is 1 at the old table}
\end{aligned}
$$

Check on the road. At 7 tables with 10 days played with the luck, the gradient of the loss equals minus Part 3's estimate at the old table, and minus the average of $\rho_t \Psi_t$ times the score one table further on, within 10^{-14}. $\nabla \rho$ equals ρ times the score within 10^{-9} (central differences).

Appendix D.2: the ratio reweights the old days

A day d has coins $G(d)$ and chance $p_w(d)$, a product over its three roads. $\mathbb{E}_{\text{old}}$ averages over old days. In line 4, Ψ_t is the exact gap of the road.

$$\mathcal{J}(w) = \sum_d p_w(d)\, G(d)$$

Part 2: the long-run average

Appendix D.2: the ratio reweights the old days

A day d has coins $G(d)$ and chance $p_w(d)$, a product over its three roads. $\mathbb{E}_{\text{old}}$ averages over old days. In line 4, Ψ_t is the exact gap of the road.

$$\begin{aligned} \mathcal{J}(w) &= \textstyle\sum_d p_w(d)\, G(d) && \textit{Part 2: the long-run average} \\ &= \textstyle\sum_d p_{\text{old}}(d)\, \dfrac{p_w(d)}{p_{\text{old}}(d)}\, G(d) && \textit{multiply and divide by } p_{old}(d) \end{aligned}$$

Appendix D.2: the ratio reweights the old days

A day d has coins $G(d)$ and chance $p_w(d)$, a product over its three roads. $\mathbb{E}_{\text{old}}$ averages over old days. In line 4, Ψ_t is the exact gap of the road.

$$
\begin{aligned}
\mathcal{J}(w) &= \textstyle\sum_d p_w(d)\, G(d) \\
&= \textstyle\sum_d p_{\text{old}}(d)\, \dfrac{p_w(d)}{p_{\text{old}}(d)}\, G(d) \\
&= \mathbb{E}_{\text{old}}\big[\rho_1\rho_2\rho_3\, G_1\big]
\end{aligned}
$$

Part 2: the long-run average

multiply and divide by $p_{old}(d)$

a day's ratio is a product of three

Appendix D.2: the ratio reweights the old days

A day d has coins $G(d)$ and chance $p_w(d)$, a product over its three roads. $\mathbb{E}_{\text{old}}$ averages over old days. In line 4, Ψ_t is the exact gap of the road.

$$
\begin{aligned}
\mathcal{J}(w) &= \textstyle\sum_d p_w(d)\, G(d) && \textit{Part 2: the long-run average} \\
&= \textstyle\sum_d p_{\text{old}}(d)\, \dfrac{p_w(d)}{p_{\text{old}}(d)}\, G(d) && \textit{multiply and divide by } p_{old}(d) \\
&= \mathbb{E}_{\text{old}}\big[\rho_1\rho_2\rho_3\, G_1\big] && \textit{a day's ratio is a product of three} \\
\mathcal{J}(w) - \mathcal{J}(w_{\text{old}}) &\approx \mathbb{E}_{\text{old}}\Big[\textstyle\sum_t \rho_t\, \Psi_t\Big] && \textit{PPO keeps one ratio a step (first order)}
\end{aligned}
$$

Appendix D.2: the ratio reweights the old days

A day d has coins $G(d)$ and chance $p_w(d)$, a product over its three roads. $\mathbb{E}_{\text{old}}$ averages over old days. In line 4, Ψ_t is the exact gap of the road.

$$
\begin{aligned}
J(w) &= \textstyle\sum_d p_w(d)\, G(d) && \textit{Part 2: the long-run average} \\
&= \textstyle\sum_d p_{\text{old}}(d)\, \dfrac{p_w(d)}{p_{\text{old}}(d)}\, G(d) && \textit{multiply and divide by } p_{old}(d) \\
&= \mathbb{E}_{\text{old}}\left[\rho_1 \rho_2 \rho_3\, G_1\right] && \textit{a day's ratio is a product of three} \\
J(w) - J(w_{\text{old}}) &\approx \mathbb{E}_{\text{old}}\left[\textstyle\sum_t \rho_t\, \Psi_t\right] && \textit{PPO keeps one ratio a step (first order)}
\end{aligned}
$$

Check on the road. By enumeration of the 216 outcomes of a day (8 paths times 27 luck combinations), lines 1 and 3 agree within 10^{-13} at three pairs of tables. With Ψ_t the exact gap of the road, the error of line 4 falls 3.9 times (3.7 to 4.0) each time the move of the knobs is halved.